📝 QUIZ QUESTION BANK · YEAR 9–10

Advanced Water Safety Quiz Question Bank

30 higher-order questions for Kahoot, Quizlet, or written assessment — covering all 5 lessons of the Year 9–10 Advanced Water Safety unit. Answers, difficulty ratings, and teaching notes included.

Year 9–10 HPE | 30 questions | 5 lesson topics | Kahoot / Quizlet ready | Victorian Curriculum 2.0

30 Total questions	**18** Multiple choice	**6** True / False	**6** Short answer
0 Easy — all Medium–Hard–Extension	**12** Medium	**12** Hard	**6** Extension

📋 How to use this question bank

All questions are pitched at Year 9–10 analytical level — Medium through Extension. Use them for:

- **Start-of-lesson warm-up** — 3–4 questions to activate prior knowledge from the previous lesson
- **End-of-lesson exit quiz** — 4–5 questions from that lesson's section
- **End-of-unit Kahoot revision** — all 30 as a full-class game
- **Written short-answer practice** — use the Extension questions as formative assessment prompts
- **Peer quiz challenge** — pairs quiz each other using the question cards

🎮 Kahoot — kahoot.com | 🃏 Quizlet — quizlet.com | 📝 Google Forms

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

✍️ Written assessment

1 Advanced Data Analysis — Victorian & Global Drowning Trends

Q1–Q6 · 6 questions

Q1 **Medium** Multiple choice

Victoria had 52 drowning deaths in 2024–25 with a population of approximately 6.9 million. What is the correct drowning rate per 100,000 population?

A. 1.13 per 100,000

✓ **B. 0.75 per 100,000**

C. 0.54 per 100,000

D. 2.1 per 100,000

Answer: B — 0.75 per 100,000
Calculation: 52 ÷ 69 (units of 100,000 in 6.9M) = 0.75. Note: Victoria's rate is actually BELOW the national rate of 1.13 per 100,000 (2025/26) — a nuance worth discussing. It does not mean all is well, given specific problem areas (CALD, inland waterways).

Q2 **Medium** Multiple choice

People from multicultural backgrounds account for 37% of Victorian drowning deaths since 2015–16, and 58% of fishing-related drowning deaths. What term best describes this pattern?

A. Under-representation

✓ **B. Over-representation**

C. Proportional representation

D. Statistical noise

Answer: B — Over-representation
LSV reports 185 multicultural drowning deaths since 2015–16 — 37% of the total, with country of birth unknown in 22% of cases. Key teaching point: this is a structural issue, not a reflection on any community's individual behaviour. Access to swim lessons, language barriers to safety messaging, and occupational risk (fishing) are structural factors.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q3 Hard Multiple choice

If Victoria drowned at the Australian national rate of 1.13 per 100,000 (2025/26), approximately how many Victorian drowning deaths per year would be expected?

A. 38 deaths

B. 52 deaths

✓ **C. 83 deaths**

D. 120 deaths

Answer: C — 83 deaths
Calculation: 1.2 × 69 (units of 100,000) = 82.8 ≈ 83. Counterintuitive insight: Victoria is already BELOW the national average — dropping to the national rate would mean MORE deaths, not fewer. This is a powerful discussion starter about statistics and perception.

Q4 Hard True / False

Victoria's non-fatal drowning rate in 2024–25 was approximately 60% above the 10-year average, suggesting the total scale of the water safety problem is much larger than fatal figures alone indicate.

True

False

Answer: True
Non-fatal drownings represent near-death experiences, long-term trauma, and significant healthcare costs. Each near-drowning also represents a failure of prevention. The 60% above average figure in 2024–25 signals a systemic problem that fatal statistics alone would understate.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q5 **Extension** Short answer

A student argues: "Victoria should be proud — our drowning rate of 0.75 per 100,000 is well below the national average of 1.2." Evaluate this argument. What does the student get right, and what important context are they missing?

Student response area — 4–6 sentences expected at B/A standard…

Sample A-standard answer: The student is correct that Victoria's overall rate is below the national average — this is accurate and worth acknowledging. However, this obscures significant internal disparities: people from multicultural backgrounds accounted for 37% of deaths since 2015–16, inland waterway deaths have been rising, and the 2023–24 total of 54 sat above the ten-year average despite being lower than the 59 recorded in 2022–23 and 61 in 2020–21. A rate below average does not mean the problem is under control — it means the baseline is already too high and particular communities are being failed by the current system.

Q6 **Hard** Multiple choice

The WHO reports around 300,000 global drowning deaths per year, with 92% occurring in low- and middle-income countries. The global drowning death rate has fallen 38% since 2000, from 6.1 to 3.8 per 100,000. What does this tell us about the relationship between income and drowning risk?

A. People in low-income countries are worse swimmers

✓ **B. Structural factors like infrastructure, access to swim lessons, and lifeguard services significantly influence drowning risk**

C. Climate differences between countries explain the disparity

D. Drowning is primarily a problem in developing nations and not relevant to Australia

Answer: B — Structural factors
This question bridges individual behaviour and structural analysis. The 90% figure reflects lack of pool fencing laws, limited lifeguard infrastructure, no swim education in schools, and higher rates of open-water occupational activities (fishing, rice cultivation). This connects directly to the social determinants theme in Lesson 3.

2 Physiology of Drowning & CPR Mechanics

Q7–Q12 · 6 questions

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q7 Medium Multiple choice

Which of the following correctly describes the physiological event called laryngospasm?

A. The heart stopping due to lack of oxygen

✓ **B. Involuntary closure of the vocal cords on contact with water, preventing water entry to the lungs**

C. The brain losing consciousness due to low blood pressure

D. Rapid loss of muscle coordination in cold water

Answer: B — Laryngospasm
Laryngospasm accounts for "dry drowning" in approximately 10–15% of cases — the person asphyxiates without water entering the lungs. It is involuntary and cannot be consciously controlled. Even in dry drowning, the outcome (cerebral hypoxia) is the same as wet drowning if CPR is not started promptly.

Q8 Medium True / False

Standard chest compressions during CPR generate approximately 25–30% of normal resting cardiac output — which is sufficient to prevent irreversible brain cell death while waiting for emergency services.

True | **False**

Answer: True
25–30% of normal output is enough to keep the brain alive in a holding pattern. This is why bystander CPR — even imperfect CPR — dramatically improves survival rates. It buys time until defibrillation or ROSC. The figure also explains why compressions should be hard and fast (100–120 bpm, 5–6cm depth) — partial effort generates even less output.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q9 Hard Multiple choice

A drowning victim receives bystander CPR starting at 3 minutes after cardiac arrest. Which of the following best explains why this produces significantly better outcomes than CPR starting at 9 minutes?

A. CPR is more effective when the body is still warm

✓ **B. Brain cells begin dying irreversibly after 4–6 minutes without oxygenated blood; starting at 3 minutes limits the window of cell death before compressions restore minimum perfusion**

C. The heart is more likely to restart spontaneously within 3 minutes

D. Emergency services always arrive faster when CPR is started early

Answer: B
The mechanism is cerebral hypoxia — irreversible neuron death begins at 4–6 minutes. Starting CPR at 3 minutes keeps minimum oxygenation going before significant cell death occurs. At 9 minutes, substantial brain damage has already occurred regardless of how good the CPR is. This is why the 4-minute window and bystander CPR training matter so much.

Q10 Hard Multiple choice

Secondary drowning involves pulmonary oedema developing hours after a submersion incident. Which of the following best describes the clinical significance of this condition?

A. It is a common complication that affects most near-drowning survivors

✓ **B. It is rare but potentially fatal — symptoms may not appear for hours after the incident, making it easy to dismiss a near-drowning as resolved**

C. It only affects people who were underwater for more than 5 minutes

D. It can be fully prevented by performing rescue breaths during CPR

Answer: B
Secondary drowning (pulmonary oedema) affects a small percentage of near-drowning victims but is potentially fatal because it develops hours later. Symptoms — coughing, fatigue, chest pain, laboured breathing — can appear up to 24 hours post-incident. The key clinical point: ANY submersion incident, even if the person appears fine, warrants medical review. Parents and coaches must be aware of this.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q11 Extension Short answer

Explain the "1-10-1" cold water survival rule. Why does the rule suggest 10 minutes of useful swimming ability rather than longer, even for a strong swimmer?

Student response area…

Sample A-standard answer: The 1-10-1 rule describes three phases: approximately 1 minute to control breathing (cold shock causes involuntary hyperventilation); approximately 10 minutes of useful swimming before peripheral vasoconstriction and muscle cooling cause swimming failure; up to 1 hour before hypothermia incapacitates (varies with water temperature, body composition, clothing). The 10-minute swimming window is limited even for strong swimmers because cold water causes neuromuscular dysfunction independent of fitness — muscles simply stop responding to nerve signals as temperature drops. A world-class swimmer cannot swim indefinitely in 12°C water.

Q12 Hard True / False

The correct sequence of physiological events during drowning is: submersion → laryngospasm → cardiac arrest → hypoxia → cerebral hypoxia.

True	False

Answer: False
The correct sequence is: submersion → laryngospasm (involuntary) → water entry / hypoxia develops → cardiac arrest → cerebral hypoxia. Cardiac arrest follows hypoxia, not the other way around. In dry drowning, laryngospasm persists and hypoxia still develops — just without water entering the lungs. Cerebral hypoxia is the terminal event that leads to brain death if CPR is not started promptly.

3 Social Determinants of Drowning Risk

Q13–Q18 · 6 questions

Q13 Medium Multiple choice

In 2024–25, males accounted for approximately 73% of Victorian drowning deaths. Which of the following best explains this disparity from a social determinants perspective?

A. Males are biologically weaker swimmers than females	✓ **B. Masculinity norms — social expectations that males should take**

risks, not back down from dares, and demonstrate strength — lead to higher engagement in risky water behaviours

C. Males spend more time near water than females

D. Males are less likely to have completed swimming lessons

Answer: B — Masculinity norms
This is a social determinant, not a biological one. Research consistently links risk-taking behaviour near water to gender norms that frame backing away from a dare as weakness. This connects to alcohol consumption, swimming at unpatrolled locations, and solo swimming. Prevention strategies targeting this group need to reframe water safety as intelligence, not weakness.

Q14 **Medium** Multiple choice

Which of the following is a structural explanation for CALD community over-representation in Victorian drowning statistics — rather than an individual explanation?

A. CALD community members make poor decisions near water

B. CALD families don't value water safety

✓ **C. Many CALD community members grew up in countries without access to swimming lessons or public pools, creating a skill gap that is structural and historical rather than individual**

D. CALD community members are more likely to swim in cold water

Answer: C — Structural explanation
Distinguishing structural from individual explanations is a core Year 9–10 skill. Structural factors exist at the system level — lack of swim infrastructure in countries of origin is not an individual choice. Other structural factors: language barriers to safety messaging, higher rates of occupational water exposure (commercial fishing), lower rates of swim lesson access due to cost and cultural norms around mixed-gender facilities.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q15 Hard True / False

The social determinants of health framework suggests that health outcomes are primarily determined by individual choices and personal responsibility.

True	False

Answer: False
The WHO Social Determinants of Health Framework explicitly states that health outcomes are shaped by "the conditions in which people are born, grow, live, work, and age" — systemic factors largely outside individual control. Individual choices exist within these structures. A person who grew up without access to swim lessons faces a structurally different risk level to someone who had unlimited pool access as a child, regardless of their individual choices as an adult.

Q16 Hard Multiple choice

A school wants to reduce drowning risk for its CALD student population. Which of the following interventions specifically targets the structural barriers — rather than just individual awareness?

A. Showing CALD students a video about drowning statistics

B. Sending home a translated safety flyer

✓ **C. Partnering with LSV to deliver free, culturally safe, in-language swim lessons in the local community, delivered by CALD community members**

D. Including a chapter on water safety in the CALD student welcome booklet

Answer: C
Options A, B, and D address awareness but not the structural barrier — which is skill gap and access. Free, in-language lessons delivered by community members addresses: cost (removes financial barrier), language (removes communication barrier), cultural safety (addresses discomfort with mixed-gender or unfamiliar environments), and skill development (directly addresses the root cause). This is the difference between individual-level and structural-level intervention.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q17 Extension Short answer

Define "health inequity" and explain, using the Victorian drowning data, why the CALD over-representation in drowning deaths is an example of health inequity rather than health inequality.

Student response area…

Sample A-standard answer: Health inequality refers to any difference in health outcomes between groups. Health inequity refers specifically to differences that are unfair, avoidable, and rooted in unjust social conditions. The over-representation of multicultural communities in Victorian drowning deaths (37% of deaths since 2015–16) is an inequity because it results from structural barriers — limited swim access in countries of origin, language barriers to safety messaging — that CALD community members did not choose and could not individually overcome. It is avoidable through targeted structural interventions (funded community swim programs, translated resources, culturally safe delivery), making it a matter of social justice, not just statistical difference.

Q18 Hard True / False

Alcohol is a factor in approximately 20–30% of adult drowning deaths in Victoria, suggesting that alcohol restriction near water is a relevant prevention strategy.

True | False

Answer: True
Alcohol impairs judgment, coordination, thermoregulation, and creates overconfidence — a dangerous combination near water. The 20–30% figure reflects its role as a contributing factor rather than the sole cause. Prevention implications: signage at popular swimming locations, community norm-setting campaigns targeting social swimming culture, and alcohol management at organised water events.

4 Prevention Strategies & International Comparison

Q19–Q24 · 6 questions

Q19 Medium Multiple choice

Which drowning prevention strategy has the strongest evidence base for reducing child drowning in home pools?

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

A. Warning signs at pool gates

B. Parent supervision during pool use

✓ **C. Four-sided pool isolation fencing**

D. Mandatory swimming lessons for children

Answer: C — Four-sided pool isolation fencing
Four-sided isolation fencing (surrounding the pool on all four sides, separate from the house) reduces child drowning in home pools by approximately 83% — the strongest evidence base of any single prevention intervention. Three-sided fencing (where one side is the house) is significantly less effective. This is why Victorian pool fencing laws mandate four-sided isolation in new builds.

Q20 Hard Multiple choice

The RLSS UK "Float to Live" campaign teaches floating as the PRIMARY response to any water emergency. How does this differ from the traditional Victorian water safety message?

A. There is no difference — Victoria already uses "Float to Live" as its primary message

✓ **B. Victorian messaging focuses primarily on swimming to safety; "Float to Live" argues that floating and conserving energy should come first, only attempting to swim when rested**

C. "Float to Live" is aimed at children only, while Victorian messaging targets all ages

D. "Float to Live" is a UK program that has no relevance to Australian water environments

Answer: B
The key distinction is physiological priority. Swimming to safety requires energy; floating conserves it. In cold water especially, the "swim immediately" instinct wastes the limited swimming window (the 10-minute rule). RLSS UK argues that floating first — controlling panic and breathing — then assessing swimming options is physiologically superior. Victorian messaging could adopt this emphasis without abandoning existing content. Source: rlss.org.uk

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q21 **Hard** True / False

Warning signs at unpatrolled beaches have the same evidence strength as lifeguard supervision for preventing adult drowning deaths.

True	**False**

Answer: False
Warning signs have limited evidence — they raise awareness but do not prevent the behaviour (swimming at unpatrolled locations) that leads to drowning. The strongest evidence is lifeguard supervision: 0 deaths at patrolled Victorian locations in 2024–25 vs 54 at unpatrolled locations. Signs are low-cost and useful as part of a broader strategy but cannot substitute for active supervision. This is a critical distinction in prevention strategy evaluation.

Q22 **Medium** Multiple choice

When evaluating which drowning prevention strategies to fund, what is the most important criterion?

A. The strategy that is cheapest to implement	B. The strategy that is most familiar to the public
✓ C. The strategy with the strongest evidence of effectiveness for the specific target population and context	D. The strategy that has been used the longest internationally

Answer: C — Evidence + target population + context
Pool fencing has the strongest overall evidence but is irrelevant to inland waterway deaths. Lifeguard patrols work at beaches but don't help at farm dams. CALD swim programs have good evidence for specific communities but limited reach. Effective prevention policy matches strategy to the specific risk profile of the target group — which is exactly what the Victorian data analysis enables.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q23 Extension Short answer

You have been asked to advise the Victorian Government on how to allocate $2 million to reduce drowning deaths. Identify two strategies you would prioritise, justify each with evidence, and explain the trade-offs involved in your choices.

Student response area…

Sample A-standard answer: Priority 1 — Expand patrolled beach hours and locations ($1.2M): strongest evidence base (0 deaths at patrolled sites 2023–24), direct intervention against the highest-risk behaviour, measurable outcome (expanded patrol hours). Trade-off: high cost per prevented death; doesn't address inland waterway or CALD risk. Priority 2 — Subsidised in-language CALD community swim programs ($800K): addresses the structural access gap behind 37% of deaths since 2015–16; demonstrated impact in RLSSA pilot programs; long-term skill development. Trade-off: slower to show measurable results; requires sustained funding beyond a single year. Together these address the two largest identified risk gaps in the Victorian data.

Q24 Hard Multiple choice

Which of the following best describes why mandatory swim lessons for adults have weaker evidence for reducing drowning than mandatory swim lessons for children?

A. Adults learn less effectively than children

✓ **B. Adult drowning is more strongly associated with risk-taking behaviour, alcohol, and structural factors than with inability to swim — so skill alone does not address the root causes**

C. Adult swim lessons are too expensive to be practical

D. There is no evidence that swim lessons work for anyone

Answer: B
Most Victorian adult drowning victims could swim to some degree. The risk factors are behavioural (alcohol, swimming alone, ignoring flags) and structural (CALD access gaps, inland waterway use without safety knowledge). Adult interventions need to address these root causes — behaviour change campaigns, structural access programs, and targeted messaging — rather than assuming skill deficit is the primary issue.

5 Assessment Task — Evidence, Argument &

Q25–Q30 · 6 questions

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Communication

Q25 **Medium** Multiple choice

A policy brief is addressed to the Victorian Minister for Sport. Which of the following best describes what this means for how the brief should be written?

A. It should be written in simple language because the minister may not know much about the topic

B. It should include as much technical detail as possible to demonstrate expertise

✓ **C. It should be concise, evidence-based, action-oriented, and frame recommendations in terms of government capacity, cost-effectiveness, and measurable outcomes**

D. It should avoid statistics because politicians prefer stories

Answer: C
Policy briefs to ministers are professional documents designed to support decision-making. They prioritise: clarity over comprehensiveness, actionable recommendations over description, evidence over opinion, and measurable outcomes over aspirational statements. Ministers receive dozens of briefs — brevity, clarity, and specificity are essential.

Q26 **Medium** True / False

In an evidence-based health communication product, citing a government health website URL is a stronger source than citing a peer-reviewed journal article.

True

False

Answer: False
Peer-reviewed journal articles undergo rigorous blind review and represent the strongest form of evidence. Government websites (e.g. lsv.com.au, who.int) are credible primary sources for statistics and policy positions. For the purposes of this unit, government and reputable NGO sources (LSV, RLSSA, WHO) are considered acceptable primary sources. Social media, general websites, and Wikipedia are not.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q27 **Hard** Multiple choice

A student designing a water safety campaign targeting young males chooses Instagram as their platform and uses the message: "Real strength is knowing when not to jump in." Which persuasion principle does this message employ?

A. Fear appeal — threatening consequences to change behaviour

✓ **B. Reframing — redefining the meaning of masculinity to align safety-seeking behaviour with the target audience's values**

C. Social proof — showing that everyone else is making safe choices

D. Authority — using an expert to endorse the message

Answer: B — Reframing
This message directly targets the masculinity norm barrier by redefining what "strength" means — rather than opposing the value (strength) it reframes it. Research shows fear appeals are often ineffective with young males and can backfire. Reframing is a more sophisticated technique that works with existing values rather than against them. This demonstrates deep understanding of the target audience.

Q28 **Hard** Multiple choice

A school audit of water safety provision finds that the school teaches beach flag identification in Year 7 but has no content addressing inland waterway safety, physiological knowledge, social determinants, or Year 9–10 content. Which Curriculum 2.0 codes are NOT being addressed?

A. VC2HP8P08 and VC2HP8P10 only

B. VC2HP8M01 only

✓ **C. VC2HP10P02, VC2HP10P08, VC2HP10P10, VC2HP10P09, and VC2HP10M01**

D. All codes are adequately addressed by the Year 7 content

Answer: C — Year 9–10 codes
VC2HP8P08/126 are Year 7–8 codes (partially addressed). VC2HP10P02/144/147/148 and VC2HP10M01 are Year 9–10 codes requiring analysis of health behaviours across diverse groups, evaluation using health data, planning community strategies, and specialised movement concepts. A school that only covers Year 7–8 beach safety content is not meeting the full Victorian Curriculum requirements for Years 9–10.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Q29 Extension Short answer

What is the difference between a measurable success metric and a vague aspiration? Give one example of each in the context of a water safety prevention campaign targeting young males.

Student response area…

Sample A-standard answer: A measurable success metric specifies a concrete, observable outcome with a timeframe — it can be evaluated. A vague aspiration describes a desired state without specifying how success would be measured or when. Vague: "We want young males to make safer choices near water." This cannot be measured — how would we know if it was achieved? Measurable: "A 15% reduction in 15–24 year old male unpatrolled location drowning deaths in Victoria within 3 years of campaign launch, measured against LSV Drowning Report data." This names the indicator, the target group, the magnitude of change, the timeframe, and the measurement source.

Q30 Hard True / False

Acknowledging counter-arguments in a policy brief weakens the argument by introducing doubt.

True	False

Answer: False

Acknowledging and responding to counter-arguments (called "refutation") strengthens credibility. A reader who is aware of the counter-argument and sees it addressed is more persuaded than one who is simply not told about it. This is a key feature of A-standard persuasive writing. Pretending a counter-argument doesn't exist reads as naive or evasive — addressing and refuting it demonstrates sophisticated thinking.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

Interleaved Retrieval Set — Years 9–10

IL1–IL12 · 12 questions · mixed topics

Why this set is different. Every other section of this bank groups questions by lesson. That is useful while a topic is being taught and close to useless afterwards, because the student always knows which knowledge the question wants — the hardest part of the job is done for them before they start. This set mixes topics deliberately and in no stated order, so the student has to work out *which* knowledge applies before retrieving it. That decision is the skill that transfers to a real waterway.

How to use it. After the unit, not during. Do not tell students which topic each question comes from, and do not reorder them into topic groups — that undoes the whole point. Expect scores to drop against the blocked sections; that drop is the measure of how much of their earlier confidence came from context rather than knowledge. VTLM 2.0: *Vary practice*, Practice 2 (interleaving) and *Revisit and review*.

IL1 Hard Discrimination INFERENCE VS DATA

Victoria's fatal drowning rate was 0.75 per 100,000 and the national rate was 1.13 per 100,000. A student concludes that Victoria's prevention programs are working better than other states'. What is the strongest objection?

A. The rates are too close to compare

✓ **B. Rate differences can reflect geography, climate and exposure, not just program quality**

C. Victoria has a larger population so its rate is always lower

D. The national figure covers a different year, so no comparison is possible

Answer: B — Confounding. States differ in coastline, inland waterways, climate and participation, all of which affect exposure independently of prevention. Option D is a real caution but weaker — the two figures are comparable in direction; the issue is what causes the difference.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

IL2 Medium Recall

What is laryngospasm, and roughly what share of drowning deaths does it account for?

A. Fluid in the lungs hours later; about 25%

✓ **B. Involuntary closure of the vocal cords; about 10–15%**

C. Cardiac arrhythmia from cold; about 40%

D. Loss of consciousness from hypoxia; about 60%

Answer: B — Involuntary closure of the vocal cords on water contact, roughly 10–15% of drowning deaths. Water never enters the lungs, and the person still asphyxiates — which is why absence of water in the lungs does not rule out drowning.

IL3 Hard Discrimination STRUCTURAL VS STEREOTYPE

Which of these is a *structural* explanation rather than a stereotype?

A. Some communities are simply not water people

✓ **B. Swimming lessons cost money and are rarely offered in community languages**

C. Young men do not listen to safety advice

D. Migrants are less confident around water

Answer: B — Only B names a condition that could be changed by a decision. The others attribute the outcome to the group itself. This distinction is the whole of the social determinants lesson and the most common place Year 9–10 answers fall down.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

IL4 Medium Recall

Standard chest compressions generate roughly what proportion of normal resting cardiac output?

A. About 5%	✓ **B. About 25–30%**
C. About 60%	D. About 90%

Answer: B — About 25–30% — enough to prevent brain death for roughly 10 to 15 minutes pending defibrillation. The point is that partial perfusion buys time; it does not restore normal function.

IL5 Hard Discrimination EVIDENCE HIERARCHY

Four prevention strategies. Which would you expect to have the *strongest* evidence base, and why?

A. A social media awareness campaign — it reaches the most people	✓ **B. Four-sided pool fencing — it works whether or not anyone is paying attention**
C. Warning signage at unpatrolled beaches — it is present at the point of risk	D. A school talk by a guest speaker — it is memorable

Answer: B — Engineering and legislation generally outperform education, which outperforms awareness, because the first group does not depend on a person making the right decision in the moment. The reasoning matters more than the ranking.

IL6 Medium Recall

Country of birth was unknown in what share of Victorian drowning cases over the past decade — and what does that do to the 37% multicultural figure?

A. 5%; it makes the figure precise	✓ **B. 22%; it makes 37% a lower bound rather than a point estimate**
C. 50%; it makes the figure unusable	D. 0%; the figure is complete

Answer: B — 22% unknown, so 37% is a floor. Stating the limitation alongside the number is what separates a Year 9–10 answer from a Year 7–8 one.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

IL7 Hard Discrimination EXPLANATION SHAPES RESPONSE

95% of diving-related spinal injury patients in Australia are male. A student writes that this shows men are biologically more reckless. What is wrong with that?

A. The statistic is incorrect

B. Nothing — it is a reasonable inference

✓ **C. It treats a socially produced pattern as a fixed biological trait, which also makes it unchangeable**

D. 95% is too high to be meaningful

Answer: C — The analytical error and the practical cost are the same thing: a biological account implies nothing can be done, while a social-norms account identifies something that can be changed. 'Feet First, First Time' addresses knowledge; the norms are what the structural explanation targets.

IL8 Medium Recall

What is secondary drowning, and what is the appropriate response to a near-miss?

A. Drowning on a second attempt at rescue; no action needed

✓ **B. Pulmonary oedema developing hours later; seek medical review after any submersion incident**

C. Cold water shock on re-entry; warm the person up

D. A second cardiac arrest; repeat CPR

Answer: B — Fluid in the lungs developing hours after the event — coughing, laboured breathing, fatigue, chest pain. Rare but potentially fatal, and the reason a person who 'seems fine' after a submersion still needs review.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

IL9 Hard Discrimination EVIDENCE QUALITY

Which question would produce the *strongest* evidence for a policy brief?

A. Do people think signage works?

✓ **B. What happened to drowning rates where four-sided fencing was made mandatory, compared with where it was not?**

C. How many people have seen the current campaign?

D. Do lifeguards believe their work is effective?

Answer: B — B compares an intervention against a counterfactual. The others measure awareness, reach or opinion — all easier to collect and none of which establish that the intervention changed outcomes.

IL10 Medium Recall

Below what water temperature does cold water shock typically begin, and how fast can it incapacitate?

A. Below 20°C; within 10 minutes

✓ **B. Below about 15°C; within about 1 minute**

C. Below 5°C; within 30 seconds

D. Below 10°C; within an hour

Answer: B — Below roughly 15°C, with incapacity possible inside a minute — involuntary gasp, hyperventilation, peripheral vasoconstriction and possible arrhythmia. Victorian inland waterways sit in this range for much of the year, including on warm days.

© 2026 Emilie Harrison · CC BY 4.0 · emilie-harrison.github.io/swim-water-safety · Third-party content excluded

IL11 Hard Discrimination DATA LITERACY

You have data showing drowning deaths rose 37% at coastal locations against baseline. Before using it in an argument, what must you check first?

A. Whether the figure is from a government source

✓ **B. Whether the comparison period is the same length and season as the baseline**

C. Whether the increase is statistically significant at $p < 0.05$

D. Whether other states had the same increase

Answer: B — Period matching. An eight-month figure set against a twelve-month baseline manufactures a change that is not there. The other checks matter, but this one invalidates the comparison outright if it fails.

IL12 Hard Discrimination TRANSFER

Which of these is the clearest example of applying knowledge from one part of the unit to a problem in another?

A. Reciting the four prevention criteria from memory

✓ **B. Using the social determinants framework to explain why a signage campaign under-performs in a multilingual suburb**

C. Listing the physiological stages of drowning in order

D. Calculating a rate per 100,000

Answer: B — Transfer is combining frameworks across lessons to explain something neither covers on its own. The other three are retrieval within one topic — necessary, but not the same skill.

© 2026 Emilie Harrison. This resource is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). You are free to copy, share, adapt and build on it — including for commercial purposes — provided you give appropriate credit and indicate any changes.

Suggested attribution: "Swim & Water Safety Teaching Hub" by Emilie Harrison, licensed CC BY 4.0. Source: emilie-harrison.github.io/swim-water-safety

Third-party material is not covered. Content owned by Life Saving Victoria, Royal Life Saving Australia, Surf Life Saving Australia, VCAA, ACARA, AIATSIS, the Australian Resuscitation Council and other organisations — including data, guidelines, logos and program names — remains the property of those owners and is used here

for educational purposes. Check directly with the owner before reproducing it elsewhere.

Data sources cited in this resource. Drowning statistics are reproduced with attribution from: Life Saving Victoria. (2025). *Victorian Drowning Report 2024–25*. Life Saving Victoria: Melbourne. Royal Life Saving Australia (2026). *National Drowning Report 2026*. RLSSA: Sydney. World Health Organization (2024). *Drowning — fact sheet*. WHO: Geneva. Surf Life Saving Australia (2026). *Summer Coastal Drowning Report 2025/26*. SLSA: Sydney. These figures remain the property of the publishing organisations. If you adapt this resource, retain these citations.